

CSCE 647 IMAGE SYNTHESIS · MODULE 10 · LECTURE NOTES

Participating Media

Removing the vacuum assumption, and what it costs.

MODULE OUTCOMES

State the volume rendering equation and its coefficients.

Explain transmittance and why it is an integral.

Sample distances in homogeneous media analytically.

Implement delta tracking for heterogeneous media and say why it is unbiased.

Explain phase functions and sample Henyey–Greenstein.

1 What a medium is

Symbol	Name	Meaning	Units
σ_a	Absorption coefficient	Probability per unit length that a photon is absorbed and converted to heat.	1/length
σ_s	Scattering coefficient	Probability per unit length that a photon is redirected without loss of energy.	1/length
σ_t	Extinction coefficient	$\sigma_a + \sigma_s$. Total probability per unit length of being removed from the current ray, by either mechanism.	1/length
σ_s/σ_t	Single-scattering albedo	Of the light removed from the ray, the fraction redirected rather than absorbed.	dimensionless

The albedo is the number that governs cost. A medium with albedo near 1 absorbs almost nothing, so light scatters many times before it escapes or reaches the camera. **A cloud has an albedo of roughly 0.9999**, and photons scatter hundreds of times inside it. That bounce count, not the cost of any individual operation, is why clouds are slow to render.

THE VACUUM ASSUMPTION IS WHAT MADE RAY TRACING CHEAP

Module 01 established that radiance is invariant along a ray in a vacuum, and Module 02 used that to make a ray a **lookup**: trace until you hit something, read the radiance leaving it, done.

In a participating medium that fails. Radiance changes continuously along the ray: energy is removed by absorption and by scattering out of the ray's direction, and added by light scattering *into* the ray from every other direction.

So a ray becomes an integral along its length — and the in-scattering term inside that integral is itself an integral over the whole sphere of directions, at every point along the ray.

This is the structural reason volumetric rendering is expensive. It is not an implementation inefficiency; the assumption that made ray tracing a lookup rather than an integral has been removed.

$$\begin{aligned} dL/ds &= -\sigma_t L + \sigma_s \int p(\omega, \omega') L(\omega') d\omega' + \sigma_a L_e \\ T(a, b) &= \exp\left(-\int_a^b \sigma_t(x) ds\right) \end{aligned}$$

Transmittance T is the fraction of radiance surviving between two points — the Beer–Lambert law. Note that the exponent is an *integral* of the extinction along the path, which is why heterogeneous media are difficult: that integral has no closed form for a general density field.

$$L = \int_0^d T(\theta, s) \sigma_s(s) L_s(s) ds + T(\theta, d) L_{\text{surface}}$$

The integral form: every point along the ray scatters some light toward the camera, attenuated by the transmittance accumulated up to that point, plus the surface at the far end attenuated by the transmittance across the whole medium. **The surface rendering equation of Module 02 is the special case where $\sigma_t = 0$ everywhere between surfaces**, so $T = 1$ and only the second term survives.

2 Homogeneous media

When σ_t is constant, the transmittance integral collapses to a product and everything becomes analytic.

$$T(s) = \exp(-\sigma_t s) \Rightarrow s = -\ln(1 - \xi) / \sigma_t, \quad \text{pdf}(s) = \sigma_t \exp(-\sigma_t s)$$

This is the inversion method from Module 06 applied to the exponential distribution. Sampling the distance to the next scattering event costs one logarithm, and it is a *perfect* importance sample of the transmittance — so the σ_t in the PDF cancels against the σ_s in the estimator, leaving the albedo. **The same pleasing cancellation as Malley's method in Module 06**, and for the same reason: the density was chosen to match a factor of the integrand exactly.

PHASE FUNCTIONS ARE THE BRDF OF A MEDIUM

A phase function $p(\omega, \omega')$ gives the probability density of scattering from one direction into another. **It is normalised to integrate to 1 over the sphere** — unlike a BRDF, which may absorb, because absorption is already accounted for separately by σ_a .

Isotropic: $p = 1/4\pi$, scattering equally in all directions. A reasonable model for thin smoke and a good default.

Henyey–Greenstein: a single parameter $g \in (-1, 1)$ controls the anisotropy. Positive g is forward scattering — clouds, skin, and most biological tissue scatter strongly forward. Negative g is backward scattering.

HG was an empirical fit to measured scattering in interstellar dust, and it is used in rendering largely because **it inverts in closed form**, making it cheap to importance sample. More accurate models exist — Mie scattering for water droplets — and are rarely worth the cost.

3 Heterogeneous media

When σ_t varies through space — a cloud, a smoke simulation, a VDB volume — the transmittance integral has no closed form and distance sampling cannot be inverted analytically.

RAY MARCHING IS BIASED

The obvious approach steps along the ray in fixed increments, accumulating optical depth and attenuating as it goes.

It is biased, and not in a way that more samples per pixel fixes. The exponential of a numerically approximated integral is not an unbiased estimate of the exponential of the true integral, and the error depends on the step size rather than the sample count. Doubling the pixel samples does not help; only reducing the step does.

It also aliases. A regular step through a structured density field produces visible banding, which is usually hidden by jittering the starting offset — converting a structured artefact into noise, which is an improvement but not a fix.

It remains widely used in real time, where the bias is acceptable and the predictable cost is worth more than correctness. That is a legitimate engineering choice. The problem is making it without realising.

```
float sample_distance(Ray r, const Medium& m, RNG& rng) {
    float t = 0.0f, sigma_bar = m.majorant; // majorant >= sigma_t everywhere
    while (true) {
        t -= logf(1.0f - rng.next()) / sigma_bar; // homogeneous step
        if (t >= r.tmax) return INFINITY; // escaped
        if (rng.next() < m.sigma_t(r.at(t)) / sigma_bar)
            return t; // real collision
        // otherwise a NULL collision: continue unchanged
    }
}
```

WHY DELTA TRACKING IS EXACTLY UNBIASED

The idea is to imagine filling the medium with **fictitious matter** — 'null' scatterers that do not interact with light at all — until the total density everywhere equals a known constant majorant $\sigma_{\blacksquare}$.

The medium is now homogeneous, so distances can be sampled analytically by §2. At each sampled collision, decide whether it was real matter or fictitious filler, with probability $\sigma_t(x)/\sigma_{\blacksquare}$. A null collision changes nothing and the walk continues.

The fictitious matter cancels exactly in expectation, because a null collision leaves the photon's direction and energy untouched. The estimator is unbiased, and — crucially — **there is no step size** and therefore no step-size parameter to tune or bias to accept.

This is an idea from neutron transport (Woodcock, 1965) that arrived in graphics decades later, and it is the standard approach now.

Method	Estimates	When to use
Delta tracking	The position of the next real scattering event.	The workhorse. Use for path continuation inside a medium.
Ratio tracking	Transmittance between two points, as a continuous value rather than a binary outcome.	Shadow rays. Delta tracking gives a binary 0-or-1 estimate of transmittance, which has enormous variance; ratio tracking multiplies a running weight instead and is far smoother.
Residual ratio tracking	Transmittance, with a homogeneous control variate subtracted out.	Dense media, where plain ratio tracking still has significant variance.
Spectral tracking	Transport in media whose coefficients vary with wavelength.	Coloured smoke, absorbing liquids, skin. Needed with Module 13's spectral rendering.

Using delta tracking for shadow rays is a common and costly mistake. It answers 'did the ray get through?' with a yes or no, which is an unbiased but extremely high-variance estimate of transmittance. Ratio tracking answers 'what fraction got through?' and is the right tool.

THE MAJORANT DETERMINES THE COST

Delta tracking requires an upper bound σ_{t} on σ_{t} throughout the medium. **Any valid bound gives a correct answer** — correctness does not depend on the bound being tight.

But a loose bound is slow. Consider a cloud that is dense in one small region and nearly empty elsewhere. With a single global majorant set by the dense core, a ray through the empty periphery takes thousands of tiny steps, almost all of them null collisions, before it escapes.

The fix is a majorant grid: a coarse spatial grid storing the local maximum of σ_{t} in each cell. Empty cells have a majorant of zero and are crossed in a single step; dense cells use a tight local bound.

This is structurally the same idea as the BVH of Module 04: a cheap conservative bound that allows the empty space — which is most of the volume — to be skipped. The pattern recurs throughout rendering.

3.1 What else volumes need

- **Next-event estimation through a medium** requires the transmittance along the shadow ray, which is itself an estimate — so every shadow ray becomes a tracking loop. This is why volumetric NEE is so much more expensive than surface NEE.
- **Equiangular sampling** is essential when a light is inside or near a medium. Sampling distance by transmittance alone concentrates samples near the ray's start, while the contribution is concentrated near the light. Sampling by angle subtended at the light, combined with transmittance sampling via MIS, is the standard solution — and it is Module 08's structure applied again.
- **Subsurface scattering is a dense medium inside a boundary.** Skin, marble, milk, and wax are all media with very high albedo enclosed by a refractive surface. Brute-force volumetric transport is correct and slow; the **diffusion approximation** exploits the fact that after many scattering events the distribution becomes smooth and can be approximated analytically, which is far cheaper and is what most production renderers use for skin.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapters 11 and 14, Volume Scattering and Light Transport II	The coefficients, phase functions, and the complete delta-tracking and ratio-tracking implementations.
Novák et al. — Monte Carlo Methods for Volumetric Light Transport (survey, free)	The definitive survey. Everything in §3, with the unbiasedness proofs and the full tracking family.
Fong et al. — Production Volume Rendering (SIGGRAPH course, free)	How volumes are actually rendered in film, including majorant grids and the practical compromises.
Wrenninge — Production Volume Rendering (book site)	Course notes and code, free. Strong on the data structures.
Disney — Cloud data set (CC0)	A production cloud volume, free to use. Your test asset for Project 2.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement homogeneous media with analytic distance sampling. Render a fog-filled Cornell box and verify the transmittance against the analytic Beer–Lambert value.
- 2 Implement isotropic and Henyey–Greenstein phase functions. Verify both with χ^2 tests. Render a medium at $g = -0.5, 0$, and 0.8 and describe the difference.
- 3 Implement ray marching for a heterogeneous medium. Plot the result against step size and demonstrate that the error does not vanish with more pixel samples.
- 4 Implement delta tracking. Confirm it matches the ray-marched result as the step size goes to zero, and that it needs no step parameter.
- 5 Implement ratio tracking for transmittance. Compare the variance of shadow rays against using delta tracking for the same purpose.
- 6 Build a majorant grid for the Disney cloud and report the reduction in null collisions against a single global majorant.
- 7 Render the Disney cloud. Report render time as a function of maximum scattering events, and identify where the image stops changing.
- 8 Implement equiangular sampling for a light inside a medium and combine it with transmittance sampling using MIS. Report the variance reduction.

- 9 Render a dense high-albedo medium inside a dielectric boundary to produce subsurface scattering. Compare against a diffusion approximation if you implement one.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define the three coefficients and the single-scattering albedo, and say which governs cost.
- 2 Why does a medium break radiance invariance, and what does that do to the cost of a ray?
- 3 Write the transmittance and explain why heterogeneous media are hard.
- 4 How is distance sampled in a homogeneous medium, and what cancels in the estimator?
- 5 How does a phase function differ from a BRDF, and why is Henyey–Greenstein popular?
- 6 Why is ray marching biased, and why does increasing pixel samples not fix it?
- 7 Explain delta tracking and say precisely why the fictitious matter does not bias the result.
- 8 When should you use ratio tracking instead of delta tracking, and why?
- 9 What does the majorant affect, and what does a majorant grid do?